TF-IDF (Term Frequency – Inverse Document Frequency)

TF-IDF tells **how important a word is in a document**, compared to all other documents.

It improves over Bag of Words by:

- reducing importance of common words (“the”, “is”, “are”)
- increasing importance of unique, meaningful words

TF (Term Frequency)

TF tells **how many times a word appears in a document**.

Formula:

TF = Number of times word appears in document/total words in that document.

IDF (Inverse Document Frequency)

IDF tells **how rare a word is across documents**.

Formula:

IDF = log(total no of documents/no of documents containing the word)

- If a word appears in **many documents**, IDF is **low** → common word → not important

- If a word appears in **few documents**, IDF is **high** → rare word → important

TF-IDF Formula

TF-IDF = TF×IDF

Example (Very Simple)

We have **3 documents**:

D1: "I love apples"
D2: "I love oranges"
D3: "Apples and oranges are fruits"

Let's compute TF-IDF for the word:

"apples" in D1

Step 1: TF of "apples" in D1

D1 = "I love apples"
Total words = 3
"apples" appears = 1 time

TF = 1/3 = 0.33

Step 2: IDF of "apples"

How many documents contain the word "apples"?

- D1 → Yes
- D2 → No
- D3 → Yes

So "apples" appears in **2 documents**.

Total documents = 3

IDF = log(3/2) ~ 0.176

Step 3: TF-IDF of "apples" in D1

TF-IDF=0.33×0.176=0.058

Word	Document	TF	IDF	TF-IDF
apples	D1	0.33	0.176	**0.058**

Interpretation

- TF-IDF(“apples”, D1) is **0.058** (small value)
- Why? Because the word **appears in many documents**, so it's not unique.

If a word appeared in **only one document**, TF-IDF would become **much higher**.

TF-IDF Advantages

✔ More meaningful than Bag-of-Words
✔ Reduces unimportant words
✔ Highlights unique keywords
✔ Works well for document search, classification

TF-IDF Disadvantages

✗ Still ignores context
✗ Cannot understand synonyms
✗ Large sparse vectors
✗ Doesn't work well for deep meaning (semantics)